COMPLEX ANALYSIS

github.com/danimalabares/stack

Contents

1. Holomorphic functions 2
2. Cauchy-Riemann equations 2
3. Integration 3
4. Cauchy integral theorem 3
5. Cauchy integral formula 4
6. Taylor series 6
7. Riemann removable singularity theorem 8
8. Zeroes of a holomorphic function 9
9. Maximum principle 10
10. Argument Principle 10
11. Holomorphic functions in several variables 11
12. Taylor series in several variables 12
13. Identity theorem in several variables 12
14. Germs of holomorphic functions 12
15. Elementary symmetric polynomials and Newton formula 14
16. Weierstrass preparation theorem 16

References 20

Current status. The basic result of complex analysis, Cauchy's integral formula 5.3.1 is very well explained: if follows from Cauchy integral theorem 4.1, which is nothing else than an application of Stokes theorem, (in this case also Green's theorem), which I do not prove in this document. After Cauchy's integral formula I prove that a function is holomorphic if and only if it is analytic (i.e. has convergent power series expansion in a neighbourhood of the point) in Theorem 6.3. I use this to prove Riemann removable singularity theorem 7.1, from which I can explain how to "factor zeroes" of a holomorphic function as in Equations 8.0.1 and 10.1.1.

The so-called Argument principle, which is presented as Exercise 10.1, follows easily from the above development. Also Morera's and Liouville's theorems 5.5 and 5.6 follow easily and their proofs are outlined. Other important results not stated so far also follow, in particular: the Identity theorem, which says that if two functions coincide in a set with an accumulation point they will coincide in their domains, the open mapping theorem, and the maximum principle.

Next in Ahlfors book is a generalization of Cauchy's formula for cycles. And after that is the Residue theorem. Next are harmonic functions, in particular the mean value theorem and Schwarz theorem. Also of interest is Riemann's open mapping theorem that any simply connected region can be mapped to the unit disk.

And after all that elliptic functions.

Then multivariable complex analysis, with a beautifully worked out proof of Weierstrass preparation theorem 16.2, culminating in the proof that the ring of holomorphic functions $\mathcal{O}_n$ is a UFD via the Gauss lemma from Number Theory **??**.

1. Holomorphic functions

Definition 1.1. A function $f : \Omega \subset \mathbb{C} \to \mathbb{C}$ is *holomorphic* at $z_0 \in \Omega$ if

$$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

exists.

This is equivalent to

$$\lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h} \tag{1.1.1}$$

where h is a complex parameter.

2. Cauchy-Riemann equations

Theorem 2.1. *Let $f : \Omega \subset \mathbb{C} \to \mathbb{C}$ be a function. Let $z = x + iy$ coordinates in $\mathbb{C}$. Define $f := u + iv$ and $z_0 = x_0 + iy_0 \in \Omega$. f is holomorphic at z_0 if and only if*

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}.$$

Proof. The idea is to use Equation **??** as definition of holomorphic function. Let's differentiate along the x axis. Let $h \in \mathbb{R}$ (this time h *is* just a real number, while in the definition of holomorphic function it is a complex parameter). We see that

$$\begin{aligned} f(z_0 + h) - f(z_0) &= \Big(u(x_0 + h, y_0) + iv(x_0 + h, y_0)\Big) - \Big(u(x_0, y_0) + iv(x_0, y_0)\Big) \\ &= \Big(u(x_0 + h, y_0) - u(x_0, y_0)\Big) + i\Big(v(x_0 + h, y_0) - v(x_0, y_0)\Big) \end{aligned}$$

Takin limit as $h \to 0$ we obtain

$$\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}. \tag{2.1.1}$$

If we differentiate along the y axis we shall obtain

$$\frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}. \tag{2.1.2}$$

Indeed, we let h be again a real number, but now we multiply by i:

$$\begin{aligned} f(z_0 + ih) - f(z_0) &= \Big(u(x_0, y_0 + ih) + iv(x_0, y_0 + ih)\Big) - \Big(u(x_0, y_0) + iv(x_0, y_0)\Big) \\ &= \Big(u(x_0, y_0 + ih) - u(x_0, y_0)\Big) + i\Big((v(x_0, y_0 + ih) - v(x_0, y_0)\Big) \ldots \end{aligned}$$

To agree with our definition we must take limit as $ih \to 0$. Thus we obtain

$$\lim_{ih \to 0} \frac{f(z + ih) - f(z)}{ih} = \frac{1}{i}\left(\frac{\partial u}{\partial y} + i\frac{\partial v}{\partial y}\right) = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}$$

To conclude observe that since f is holomorphic, The expressions 2.1.1 and 2.1.2 must both equal the derivative of f at z_0. Matching real and imaginary parts gives the result.

For the converse first observe that if u and v are two real-valued functions of two real variables satisfying Cauchy-Riemann equations **??**, then they are harmonic, i.e.

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\Delta v = \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0.$$

Now if u and v have continuous first-order partial derivatives, then we know (?) that

$$u(x+h, y+k) - u(x,y) = \frac{\partial u}{\partial x} h + \frac{\partial u}{\partial y} k + \varepsilon_1$$

$$v(x+h, y+k) - v(x,y) = \frac{\partial v}{\partial x} h + \frac{\partial v}{\partial y} k + \varepsilon_2.$$

For real numbers h, k, ε_1 and ε_2, where $\varepsilon_1, \varepsilon_2$ tend to zero more rapidly than $h+ik$ in the sense that $\varepsilon_1/(h+ik) \to 0$ and $\varepsilon_2/(h+ik) \to 0$ for $h+ik \to 0$. Then we compute that

$$\begin{aligned} f(z+h+ik) - f(z) &= u(z+h+ik) + iv(z+h+ik) - (u(z) + iv(z)) \\ &= u(x+h, y+k) + iv(x+h, y+k) - u(x,y) - iv(x,y) \\ &= u_x h + u_y k + \varepsilon_1 + i(v_x h + v_y k + \varepsilon_2) \\ &= h(u_x + iv_x) + k(-v_x + iu_x) + \varepsilon_1 + i\varepsilon_2 \qquad \text{by Cauchy-Riemann} \\ &= h(u_x + iv_x) + ik(iv_x + u_x) + \varepsilon_1 + i\varepsilon_2. \end{aligned}$$

$= h(u_x + iv_x) + k(u_y +$

Then we see that

$$\lim_{h+ik \to 0} \frac{f(z+h+ik) - f(z)}{h+ik} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x},$$

so that the limit exists and thus f is holomorphic. □

3. Integration

The complex integral may first be defined for a complex-valued function f defined on a real integral as

$$\int_a^b f = \int_a^b u + i \int_a^b v.$$

And then for curves as

$$\int_\gamma f := \int_a^b f(\gamma(t))\gamma'(t)dt.$$

4. Cauchy integral theorem

The upshot about complex integration (and complex analysis, really) is that the integral $\int_\Delta f dz$ vanishes by Stokes theorem when f is holomorphic. This leads to Cauchy integral formula and several other results. Beware: this works only when the domain is simply connected.

Theorem 4.1 (Cauchy)**.** *If $f(z)$ is analytic in an open disk Δ, then*

$$\int_\gamma f(z)dz = 0 \tag{4.1.1}$$

for every closed curve γ in Δ.

Proof. The following proof comes from wiki. First write

$$\int_\gamma f(z)dz = \int_\gamma (u+iv)(dx+idy) = \int_\gamma (udx - vdy) + i\int_\gamma (vdx + udy).$$

Now apply Stokes theorem. For example, the second integral is

$$\int_{\partial\Delta} (vdx + udy) = \int_\Delta d(vdx + udy) = \int_\Delta (\partial_x u - \partial_y v)\mathrm{Vol}.$$

There's Cauchy-Riemann equations! □

The following result is essential for proving Cauchy integral formula. We are interested in taking a point away from the domain and seeing if we can extend the function holomorphically. The condition we need for this to work is given by the following theorem:

Theorem 4.2. *Let f be analytic in the region Δ' by omitting a finite number of points z_j from an open disk Δ. If f satisfies that*

$$\lim_{z\to z_j}(z - z_j)f(z) = 0 \tag{4.2.1}$$

for all j, then $\int_\gamma f dz = 0$ for any closed curve γ in Δ'.

Proof. So I think it might be using logarithmic derivative. The limit above allows to write

$$|f(z)| \leq \frac{\varepsilon}{|z - z_j|}$$

Then integrate. On the left we can bound the integral of f after applying integral triangle inequality, and on the right we have logarithmic derivative of $z - z_j$. This will be a fixed number (the index, see Definition 5.2), so that we have effectively bounded the integral. □

5. Cauchy integral formula

The way to Cauchy's integral formula is basically: holomorphic functions satisfy Cauchy-Riemann equations, which give Cauchy's integral theorem (which is just a consequence Stokes) and then apply this to the function $\frac{f(z)-f(z_0)}{z-z_0}$.

The theorem says that we can compute the value of a holomorphic function at a point as an integral around the point. The crucial point here is that the function inside the integral has a singularity at the point because of the fraction $\frac{1}{z-z_0}$. This is to be understood as the fact that the domain of holomorphy is not simply connected, and thus (we cannot apply Stokes) the integral does not vanish — it gives the value of the function at the point!

To prove a general version we shall use the notion of index of a curve about a point, which tells us how many times the curve winds about the point.

Lemma 5.1. *If the piecewise differentiable closed curve γ does not pass through the point a, then the value of the integral*

$$\int_\gamma \frac{dz}{z-a}$$

is a multple of $2\pi i$.

An idea (though not at proof) for a proof is the following: We are tempted to simply write the integrand as the logarithmic derivative of the function $f(z) = z-a$. But this isn't quite right, we must be careful with the domain of the logarithm. But it is instructive to see the computation:

$$\int_\gamma \frac{dz}{z-a} = \int_\gamma d\log(z-a) = \int_\gamma d\log|z-a| + i\int_\gamma d\arg(z-a)$$

If γ is closed then $\log|z-a|$ would return to its initial value and $\arg(z-a)$ increases or decreases by a multiple of 2π. The actual proof by Ahlfors is different.

Proof for circle. Still, the proof for a circle about a point z_0 is also trivial: this is just the case of the integral $\int_{\partial\Delta} \frac{dz}{z-z_0}$, but the curve here is $\gamma(t) = e^{it} + z_0$ so that when substituting in the parametrization we obtain $\int_0^{2\pi} \frac{ie^{it}}{(e^{it}+z_0)-z_0} = 2\pi i$. □

Definition 5.2. The *index* of the point a with respect to the closed curve γ is

$$n(\gamma, a) := \frac{1}{2\pi i}\int_\gamma \frac{dz}{z-a} \tag{5.2.1}$$

Theorem 5.3 (Cauchy Integral Formula)**.** *Suppose that $f(z)$ is analytic in an open disk Δ, and let γ be a closed curve in Δ. For any point not on γ,*

$$n(\gamma, a)f(a) = \frac{1}{2\pi i}\int_\gamma \frac{f(z)dz}{z-a} \tag{5.3.1}$$

where $n(\gamma, a)$ is the index of a with respect to γ.

Proof. This is a simple application of Theorem 4.2 for the function

$$F(z) := \frac{f(z)-f(a)}{z-a}$$

Notice the limit condition holds, so that the integral vanishes! □

Notice that we can differentiate Cauchy integral formula to obtain

$$f'(z) = \frac{1}{2\pi i}\int_\gamma \frac{f(\zeta)}{(\zeta-z)^2}d\zeta \tag{5.3.2}$$

and more generally

$$f^{(n)}(z) = \frac{n!}{2\pi i}\int_\gamma \frac{f(\zeta)}{(\zeta-z)^{n+1}}$$

Though we can just say that integrals can be differentiating by splitting into real and imaginary part, and using the real derivative, a technical lemma is used by Ahlfors to confirm that indeed we can differentiate under the integral sign.

Lemma 5.4. *Suppose that $\varphi(\zeta)$ is continuous on the arc γ. Then the function*

$$F_n(z) = \int_\gamma \frac{\varphi(\zeta)}{(\zeta-z)^n}d\zeta$$

is analytic in each of the regions determined by γ, and its derivative is $F_n'(z) = nF_{n+1}(z)$.

The importance of this result is that it shows that a holomorphic function has derivatives of all degrees.

Then we state two interesting results:

Theorem 5.5 (Morera)**.** *If $f(z)$ is defined and continuous in a region Ω, and if $\int_\gamma f dz = 0$ for all closed curves γ in Ω, then $f(z)$ is analytic in Ω.*

This is what Lee calls a *conservative covector field*, i.e. a form whose integral on closed curves vanishes. The proof of Morera's theorem then reduces to the fact that a covector field is conservative if and only if it is exact [Lee12, Theorem 11.42]. The reverse implication is the fundamental theorem of calculus. The forward implication is not very straightforward in [Lee12].

As a final remark, I add that ([Lee12, Propoistion 11.40]) a smooth covector field is conservative if and only if its line integrals are path-independent, in the sense that integrals coincide along two *piecewise* smooth curve segments with the same starting and ending points. That is, form is exact implies integral independent of homotopy class?

We finish with

Theorem 5.6 (Liouville)**.** *A bounded holomorphic function defined on all of $\mathbb{C}$ must be constant.*

Proof. Let γ be a circle of radius r about z, then by the first derivative of Cauchy formula (Eq. 5.3.2),

$$|f'(z)| \leq \frac{M}{2\pi i} \int_\gamma \frac{d\zeta}{|\zeta - z|^2}$$

and compute the integral, which is $\frac{2\pi i}{r}$. □

A fun application of this is proving the fundamental theorem of algebra. If a complex polynomial had no zeroes, $1/P(z)$ would be analytic in all of $\mathbb{C}$, and since $P(z)$ tends to ∞ as z tends to ∞, $1/P(z)$ is bounded (use Riemann sphere argument).

6. Taylor series

In this section I put wiki's proof that a function is holomorphic if and only if it is analytic. This gives for free the Taylor expansion. See Remark 8.2 to understand why this is here and not elsewhere in this notes.

Before proving the statement I need the following tool, which I also borrow from wiki.

Theorem 6.1 (Weierstrass M-test)**.** *Suppose that (f_n) is a sequence of real or complex-valued functions defined on a set A, and that there is a sequence of non-negative numbers (M_n) satisfying*

- *$|f_n(x)| < M_n$ for all $n \geq 1$ and all $x \in A$, and*
- *$\sum_{n=1}^\infty M_n$ converges.*

Then the series $\sum_{n=1}^\infty f_n(x)$ converges absolutely and uniformly on A.

Proof. The sequence S_n of partial sums is shown to be a Cauchy sequence, which must converge by completeness of $\mathbb{C}$. Indeed, for any $x \in A$ and large enough $m > n > N$, we must have

$$|S_m(x) - S_n(x)| = |\sum_{k=n+1}^m f_k(x)| \leq \sum_{k=n+1}^m |f_k(x)| < \sum_{k=n+1}^m M_k.$$

To conclude we basically apply the argument backwards: saying that $\sum_k M_k$ converges is the same as saying that the sequence of partial sums converges, which is equivalent to saying it is a Cauchy sequence, and that's just saying that the rightmost number on the right is smaller than any $\varepsilon > 0$ for suitable N. (This argument is commonly known as Cauchy convergence test.)

Notice that convergence is uniform since the limit converges regardless of the choice of x. (Of course this not very profound since the bounds M_n are uniform by hypothesis.) $\square$

Definition 6.2. A function $f : \Omega \subset \mathbb{C} \to \mathbb{C}$ is *analytic* at $z_0 \in \Omega$ if in some open disk centered at z_0 it can be expanded as a convergent power series

$$f(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n.$$

Theorem 6.3. *A function $f : \Omega \subset \mathbb{C} \to \mathbb{C}$ is holomorphic in the sense of Definition 1.1 if and only if it is analytic.*

Proof. The backward implication is easy. Suppose that $f(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n$. Then

$$\begin{aligned}
\lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} &= \lim_{z \to z_0} \frac{\sum_{n=0}^{\infty} c_n (z - z_0)^n - c_0}{z - z_0} \\
&= \lim_{z \to z_0} \frac{\sum_{n=1}^{\infty} c_n (z - z_0)^n}{z - z_0} \\
&= \lim_{z \to z_0} \sum_{n=1}^{\infty} c_n (z - z_0)^{n-1} = c_1.
\end{aligned}$$

The forward implication uses Cauchy integral formula 5.3.1 and the geometric series. Let us prove the geometric series, namely that $\lim_{n \to \infty} r^n = \frac{1}{1-r}$ for any complex number r of norm strictly less than 1:

$$\begin{array}{lccccccccccccc}
S_n & 1 & + & r & + & r^2 & + & \cdots & + & r^{n-2} & + & r^{n-1} & & \\
rS_n & & & r & + & r^2 & + & \cdots & + & r^{n-2} & + & r^{n-1} & + & r^n \\
S_n - rS_n & 1 & & & & & & & & & & & - & r^n
\end{array}$$

Thus $S_n(1 - r) = 1 - r^n$, so that $S_n = \frac{1-r^n}{1-r}$. Taking limit as $n \to \infty$ gives the result.

To prove f is analytic at $z_0 \in \Omega$ and suppose that γ is a circle contained in Ω centered in z_0. We need to construct a series expansion for every z inside the

interior of γ. By Cauchy integral formula 5.3.1,

$$\begin{aligned} f(z) &= \frac{1}{2\pi\sqrt{-1}}\int_\gamma \frac{f(\zeta)}{\zeta - z}d\zeta \\ &= \frac{1}{2\pi\sqrt{-1}}\int_\gamma \frac{f(\zeta)}{(\zeta - z_0) - (z - z_0)}d\zeta \\ &= \frac{1}{2\pi\sqrt{-1}}\int_\gamma \frac{1}{\zeta - z_0}\sum_{n=0}^{\infty}\left(\frac{z - z_0}{\zeta - z_0}\right)^n f(\zeta)d\zeta \\ &= \sum_{n=0}^{\infty}\frac{1}{2\pi\sqrt{-1}}\int_\gamma \frac{(z - z_0)^n}{(\zeta - z_0)^{n+1}}f(\zeta)d\zeta. \end{aligned}$$

The critical step in the above computation is interchanging the integral sign and the series in the last step. To make sure we can do this all we need to do is check that the last series converges. This is done via Weierstrass M-test, i.e. Theorem 6.1. We need to find uniform bounds for every term in the series. This is clear since f is bounded on γ (by compactness of γ), and the modulus of the quotient $(z - z_0)/(\zeta - z_0)$ is strictly smaller than 1 (since ζ is farther from z_0 than z), giving for every n

$$\begin{aligned} \left|\frac{1}{2\pi\sqrt{-1}}\int_\gamma \frac{(z - z_0)^n}{(\zeta - z_0)^{n+1}}f(z)d\zeta\right| &= \frac{1}{2\pi}\left|\int_\gamma \left(\frac{z - z_0}{\zeta - z_0}\right)^n \frac{f(z)}{\zeta - z_0}d\zeta\right| \\ &< \frac{1}{2\pi}\int_\gamma |K|^n \left|\frac{M}{r}\right| \\ &= |M|\,|K|^n. \end{aligned}$$

where $M = \max_{\zeta\in\gamma} f(\zeta)$, $\frac{z - z_0}{\zeta - z_0} = K < 1$ and r is the radius of γ. □

7. Riemann removable singularity theorem

The upshot about the condition that $\lim_{z\to z_0}(z - z_0)f(z) = 0$ is that this implies that $\int_\gamma f = 0$. But this actually doesn't work in the following proof, since we don't get much from that integral vanishing — that's just to say that [Ahl79, Chapter 4, Theorem 7] is confusing. Instead, I'll put here this great Wikipedia) proof:

Theorem 7.1 (Riemann's removable singularity theorem)**.** *Let $f(z)$ be analytic in a region Ω' obtained by omitting a point z_0 from a region Ω. There exists an analytic function defined on all Ω that coincides with f on Ω' if and only if $\lim_{z\to z_0}(z - z_0)f(z) = 0$. Such a function is unique.*

Proof. The direct implication is obvious since we can replace f by its holomorphic extension in the limit, which will converge to a fixed value, so that the limit is zero. Uniqueness also is trivial by continuity of any holomorphic extension.

For the converse define

$$h(z) = \begin{cases} (z - z_0)^2 f(z) & z \neq z_0 \\ 0 & z = z_0. \end{cases}$$

Clearly h is holomorphic on $\Omega \setminus \{z_0\}$, and at z_0 it's also complex-differentiable:

$$h'(z_0) = \lim_{z\to z_0}\frac{(z - z_0)^2 f(z) - 0}{z - z_0} = \lim_{z\to z_0}(z - z_0)f(z) = 0.$$

by hypothesis. That is, h is holomorphic and thus has a Taylor series about z_0 (justify! wiki):

$$h(z) = c_0 + c_1(z - z_0) + c_2(z - z_0)^2 + c_3(z - z_0)^3 + \dots$$

We have $c_0 = h(z_0) = 0$ and $c_1 = h'(z_0) = 0$, and thus

$$h(z) = c_2(z - z_0)^2 + c_3(z - z_0)^3 + \dots.$$

Hence, where $z \neq z_0$ we have

$$f(z) = \frac{h(z)}{(z - z_0)^2} = c_2 + c_3(z - z_0) + \dots.$$

However

$$g(z) = c_2 + c_3(z - z_0) + \dots$$

is holomorphic on all Ω, and is thus an extension of f. □

8. Zeroes of a holomorphic function

Perhaps the best way to remember this section is by the formula

$$f(z) = f_n(z)(z - z_0)^n, \tag{8.0.1}$$

which holds for a holomorphic function with a *zero of order* k on z_0, meaning that f and its derivatives $f^{(k)}(z_0)$ vanish for $k < n$ (see [Ahl79, p. 126]). Here f_n is a holomorphic function which does not vanish at z_0 . This fact shows that the zero z_0 is isolated, since $f_n(z)$ is continuous and nonzero in z_0, and $(z - z_0)^n$ is also nonzero in a pointed neighbourhood of z_0.

Let's explain why Equation 8.0.1 holds. Recall the function that we used to prove Cauchy integral formula (Theorem 5.3):

$$F(z) = \frac{f(z) - f(z_0)}{z - z_0}, \tag{8.0.2}$$

which has $\lim_{z \to z_0}(z - z_0)F(z) = 0$. Then we apply Riemann's removable singularity theorem 7.1 to find a holomorphic function f_1 extending F. Taking the limit as $z \to z_0$, we see that in fact $f_1(z_0) = f'(z_0)$. Substituting F by f_1 in Equation 8.0.2 we obtain

$$f(z) = f(z_0) + f_1(z)(z - z_0). \tag{8.0.3}$$

If $f(z_0)$ and $f'(z_0) \neq 0$, then we say that f has a zero of order 0 at z_0, which is one case of Equation 8.0.1. If $f'(z_0)$ also vanishes, we repeat process for f_1, and for f_2 if necessary, and so on, until we arrive at Equation 8.0.1.

Let us now stop assuming that z_0 is a zero of f and just apply the argument to f_1. We obtain

$$f_1(z) = f_1(z_0) + f_2(z)(z - z_0)$$

for some function f_2 such that $f_2(z_0) = f''(z_0)$. Substituing back in Equation 8.0.3 we obtain

$$f(z) = f(z_0) + f_1(z)(z - z_0) + f_2(z)(z - z_0)^2.$$

Repeating this process yields:

Lemma 8.1 (Finite Taylor expansion)**.** *If f is analytic in a connected open set $\Omega \subset \mathbb{C}$ and $z_0 \in \Omega$, then we can write*

(8.1.1)

$$f(z) = f(z_0)+f'(z_0)(z-z_0)+\frac{f''(z_0)}{2!}(z-z_0)^2+\ldots+\frac{f^{(n-1)}(z_0)}{(n-1)!}(z-z_0)^{n-1}+f_n(z)(z-z_0)^n$$

for some function f_n analytic on Ω.

The coefficients $\frac{f^{(n)}(z_0)}{n!}$ may be found by induction differentiating Equation 8.1.1 n times and evaluating at z_0: any term with factor $(z - z_0)$ will vanish at z_0, and we are left with $f^{(n)}(z_0) = n!f_n(z_0)$.

Remark 8.2. Now I will briefly discuss [Ahl79, Section 1.2, Chapter 5]. Since the proof presented by Ahlfors of the Riemann removable singularity theorem 7.1 did not convince me, I went to wiki and this forced me to do Taylor series before Riemann removable singularity. Of course if I ever understand Ahlfors' idea I could put Taylor after Riemann removable singularity, and after zeroes of holomorphic functions which seems much more natural

Actually this part is quite later in Ahlfors book because it requires the theory of convergence of series. To pass from the finite Taylor expansion to the infinite version we notice that the term f_{n+1} in Equation 8.1.1 is of the form

$$f_{n+1}(z) = \frac{1}{2\pi i}\int_\gamma \frac{f(\zeta)d\zeta}{(\zeta - z_0)^{n+1}(\zeta - z)}$$

where γ is a curve of radius ρ. Then "we obtain at once"

$$|f_{n+1}(z)(z - z_0)^{n+1}| \leq \frac{M|z - z_0|^{n+1}}{\rho^n(\rho - |z - z_0|}.$$

Then we must observe that the bound converges uniformly to zero in every disk of radius smaller than ρ, and by Weierstrass theorem there should be a limit function identical to zero.

Here's an important result from wiki: a complex-valued function is holomorphic if and only if it has a Taylor series expansion at every point (raidus of convergence is finite).

9. Maximum principle

Theorem 9.1 (Maximum principle)**.** *If f is analytic and non constant in a region Ω, then its absolute value $|f|$ has no maximum in Ω.*

Now look at [**?**, p. 166]: "the maximum principle for analytic functions follows from the maximum principle for harmonic functions by applying the latter to $\log|f|$ which is harmonic when $f(z) \neq 0$".

Theorem 9.2 (Maxmimum principle for harmonic functions)**.** *A nonconstant harmonic function has neither a maximum nor a minimum in its region of definition. Consequently, the maximum and the minimum on a closed bounded set E are taken on the boundary of E.*

Which basically says the same thing but now for harmonic functions and now also for the minimum. "It applies also to the minimum for the reason that $-u$ is harmonic together with u".

10. Argument Principle

A simple version of the Argument Principle does not need Residue theorem:

Exercise 10.1. Let f be a holomorphic function on a disk, non-zero on $\partial\Delta$, and let $S_k(f) := \frac{1}{2\pi\sqrt{-1}}\int_{\partial\Delta}\frac{f'}{f}z^k dz$. Prove the $S_k(f) = \sum d_i\alpha_i^k$ where α_i are all zeroes of f and d_i their multiplicities.

Proof. We use Equation 8.0.1 to write $f(z) = (z - z_1)h_1(z)$ where z_1 is a zero of f. Applying that to h_1 for another zero z_2 of f, and continuing in this way, we obtain

$$f(z) = (z - z_1)(z - z_2)\dots(z - z_n)h(z). \tag{10.1.1}$$

Computing the quotient we obtain

$$\frac{f'(z)}{f(z)} = \frac{\text{derivative of } (z - z_1)\dots(z - z_n)}{(z - z_1)\dots(z - z_n)} + \frac{h'(z)}{h(z)}$$

The right hand term will vanish upon integration since it is a holomorphic function (with no poles because $h(z) \neq 0$). The left-hand term will give a sum of $\frac{1}{z-z_i}$ upon differentiation. As it is, the result of integration is the sum of the orders of each zero i.e. the number of zeroes counted without multiplicity (because the integrals are all 1, and there's one for each zero without multiplicity).

Multiplying by z^k yields the desired result by Cauchy formula 5.3.1. More exactly, multiplying z^k in the formula gives instead of a sum of integrals $\int\frac{1}{z-z_i}$, a sum of integrals $\int\frac{z^k}{z-z_i}$ which by Cauchy formula give z_i^k (disregarding the factor $2\pi\sqrt{-1}$). □

Theorem 10.2 (Argument Principle)**.** *If f is meromorphic in Ω with zeros a_j and poles b_k, then*

$$\frac{1}{2\pi i}\int_\gamma \frac{f'(\zeta)}{f(\zeta)}d\zeta = \sum_i n(\gamma, a_i) - \sum_k n(\gamma, b_k) \tag{10.2.1}$$

Lemma 10.3 (Rouché theorem)**.** *Let γ be homologous to zero in Ω and such that $n(\gamma, z)$ is either 0 or 1 for any point z not on γ. Suppose that f and g are analytic in Ω and satisfy that $|f - g| < |f|$ on γ. Then f and g have the same number of zeros enclosed by γ.*

Compare with Misha's version

Theorem 10.4 (Rouché theorem)**.** *Let f_t be a family of holomorphic functions on a disk Δ, continuously depending on a parameter $t \in \mathbb{R}$ and non-zero everywhere on its boundary $\partial\Delta$. Prove that the number of zeros of f_t in Δ is constant.*

Proof. Consider the map $t \mapsto f_t \mapsto S(f_t)$, which is continuous by hypothesis and because S, being an integral, is continuous. Then we obtain a continuous map $\mathbb{R} \to \mathbb{R}$ with integer values, meaning it must be constant. □

A similar argument may be used to prove that a holomorphic function F defined on $\Delta \times \Delta$ gives a holomorphic map

$$y_0 \mapsto \int_{\partial\Delta}\frac{F'(x, y_0)}{F(x, y_0)}\phi(x)dx = \sum_i d_i\phi(\alpha_i) \tag{10.4.1}$$

for any holomorphic function $\phi : \Delta \to \mathbb{C}$.

11. Holomorphic functions in several variables

Lemma 11.1. [Lee24], *Theorem 1.21. Let $U \subseteq \mathbb{C}^n$ be open and $f : U \to \mathbb{C}$. The following are equivalent:*

(1) *f is holomorphic (i.e. it is continuous and has a complex partial derivative with respect to each variable at each point of U)*
(2) *f is smooth and satisfies the following Cauchy-Riemann equations:*

$$\frac{\partial u}{\partial x^j} = \frac{\partial v}{\partial y^j}, \qquad \frac{\partial u}{\partial y^j} = -\frac{\partial v}{\partial x^j} \tag{11.1.1}$$

where $z^j = x^j + \sqrt{-1}y^j$ and $f(s) = u(z) + \sqrt{-1}v(x)$.
(3) *For each $p = (p^1, \dots, p^n) \in U$ there exists a neighbourhood of p in U on which f is equal to the sum of an absolutely convergent power series of the form*

$$f(z) = \sum_{i_1,\dots,i_n} a_{i_1,\dots,i_n}(z^1 - p^1)\dots(z^n - p^n) \tag{11.1.2}$$

Proof. I will prove that if f is holomorphic then it has a Taylor series for $n = 2$. First apply Cauchy integral formula on each variable to obtain

$$f(z^1, z^2) = \frac{1}{(2\pi\sqrt{-1})^2} \int_{\substack{|z^1-w^1|=r \\ |z^2-w^2|=r}} \frac{f(w^1, w^2)}{(w^1 - z^1)(w^2 - z^2)} dw^1 dw^2$$

Now observe:

$$\frac{1}{w^1 - z^1} = \frac{1}{w^1 - p^1 + p^1 - z^1} = \frac{1}{w^1 - p^1} \frac{1}{1 - \frac{p^1 - z^1}{w^1 - p^1}} \tag{11.1.3}$$

And on the right-hand-side we have a geometric series so that we may write

$$\frac{1}{w^1 - z^1} = \frac{1}{w^1 - p^1} \sum_{k=0}^{\infty} \left(\frac{p^1 - z^1}{w^1 - p^1} \right)^k$$

finally substituting this into (11.1.3) we may take the products $(p^1 - z^1)^{k_1}(p^2 - z^2)^{k_2}$ out of the integral and define the remaining term as $a_{k_1 k_2}$. □

12. Taylor series in several variables

13. Identity theorem in several variables

Theorem 13.1 (Identity theorem)**.** *If two holomorphic functions $f, g : \Omega \subset \mathbb{C}^n \to \mathbb{C}$ coincide in an open subset of the connected open set Ω, then they coincide in all of Ω.*

Proof. Let U be the set where the function $h := f - g$ and all its partial derivatives vanish. Then U is open since for every point in U there is a neighbourhood where h is expressed as a Taylor series, whose coefficients must be given by the partial derivatives of h. By definition of U, we see that h must be zero in such a neighbourhood. U is also closed by continuity of partial derivatives of all orders. □

14. Germs of holomorphic functions

Definition 14.1. Let $U, U' \subset \mathbb{C}^n$ be neighbourhoods of 0 and $f \in \mathcal{O}_U$, $f' \in \mathcal{O}_{U'}$ holomorphic functions. We say that f and f' have the same germ, $f \sim f'$, if $f|_{U\cap U'} = f'|_{U\cap U'}$. Clearly (?), $\sim$ gives an equivalence relation on the set of pairs $(U \ni 0, f \in \mathcal{O}_U)$. An equivalence class is called **germ of a holomorphic function**. The space of germs in 0 of holomorphic functions on $\mathbb{C}^n$ is denoted $\mathcal{O}_n$.

Exercise 14.2. Prova that the ring $\mathcal{O}_n$ of germs of holomorphic functions is not finitely generated over $\mathbb{C}$ for any $n > 0$.

Proof. I think the existence of e^z as a holomorphic function satisfying the differential equation $f' = f$ is enough to show that the coefficients of its Taylor polynomial are all nonzero. This argument works for several variables as well. □

Definition 14.3. A *formal power series* in the variables $t_1, \dots, t_n$ is a sum $\sum_{i=0}^{\infty} P_i(t_1, \dots, t_n)$ where P_i are homogeneous polynomials of degree i. Addition of power series is defined componentwise, and multiplication is defined via

$$\left(\sum_{i=0}^{\infty} P_i(t_1, \dots, t_n)\right)\left(\sum_{i=0}^{\infty} Q_i(t_1, \dots, t_n)\right) = \sum_{i=0}^{\infty} R_i(t_1, \dots, t_n)$$

where $R_d(t_1, \dots, t_n) = \sum_{i+j=d} P_i(t_1, \dots, t_n)Q_j(t_1, \dots, t_n)$.

We can think of germs of functions in $\mathcal{O}_n$ as elements in the ring of power series $\mathbb{C}[\![t_1, \dots, t_n]\!]$. I think there is no problem to prove this statement, nor the fact that power series is actually a ring with units the nonzero constants and zero the zero constant.

Exercise 14.4. Prove that $\mathcal{O}_n$ has no zero divisors.

Proof. Suppose that $PQ = 0$ but neither of P nor Q are zero. Then $P_i \neq 0$ for some i and $Q_j \neq 0$ for some j. Then we can write

$$P_iQ_j = -\sum_{\substack{p+q=i+j \\ p\neq i, q\neq j}} P_pQ_q$$

And then what. Other way is by induction. For $n = 0$ we are the complex numbers so no problem. Suppose that $\mathcal{O}_n$ has no zero divisors. Then it looks like we can deal with degrees smaller than $n + 1$, but the d-th term is not a simple product of P_iQ_j with $i + j = d$, but a sum. So not sure too. □

Definition 14.5. A ring R is called *local* if it contains an ideal $I \subset R$ such that all elements $r \notin I$ are invertible.

It is an easy exercise to show that this definition is equivalent to having a unique maximal ideal.

Exercise 14.6. Prove that the ring $\mathbb{C}[\![t_1, \dots, t_n]\!]$ is not finitely generated over $\mathcal{O}_n \subset \mathbb{C}[\![t_1, \dots, t_n]\!]$.

Proof. I thought that $\mathcal{O}_n$ would be the same as $\mathbb{C}[\![t_1, .., t_n]\!]$... the natural map is surely injective but why not surjective? There are power series that are not holomorphic functions? Maybe because of radius of convergence? No, because germs of holomorphic functions can be defined very near the origin... Every power series has a nonzero radius of convergence, right? □

Now I will discuss zeroes of holomorphic functions of several variables.

Definition 14.7. Let $f \in \mathcal{O}_n$ be a germ of holomorphic function on $\mathbb{C}^n$. Write its Taylor series $f(z) = \sum_{i=0}^{\infty} P_i(t_1, \ldots, t_n)$, where P_i are homogeneous polynomials of degree i. We say that f has a *zero of order (or multiplicity) k in* 0 if $P_0 = \ldots = P_{k-1} = 0$. In this situation *principal part* of the function f is the homogeneous polynomial P_k.

The following exercise shows that a holomorphic change of coordinates will preserve the order of a zero, and the principal part of the new function will be determined by the differential of the change of coordinate map. We will need to change coordinates when we do Weierstrass Preparation theorem **??**.

Exercise 14.8. Let $\Phi(t_1, \ldots, t_n) = (F_1(t_1, \ldots, t_n), \ldots, F_n(t_1, \ldots, t_n))$ be the holomorphic coordinate change around zero (?), with $F_i(0, \ldots, 0) = 0$ and $A := \left(\frac{\partial F_i}{\partial t_j}\right)_0$ its differential (I suppose $\det A \neq 0$. Prove that

(1) For any germ $f \in \mathcal{O}_n$ which has multiplicity k, the function $\Phi^*(f)$ has zero of the same multiplicity.
(2) The principal part of $\Phi^*(f)$ is obtained from the principal part of f by action of A.

Sketch of proof. (1) The condition that $\det A \neq 0$ implies that all F_i must have linear term—otherwise their partial derivatives would vanish when evaluated at zero. When substituting $f(z_1, \ldots, z_n) = \sum_{|\alpha| \geq k} \alpha z^\alpha$ with Φ we find that there must be a term of order k.
(2) The case for $n = 1$ is clear. The general case follows, I think, from the observation that the derivative A recovers the linear terms, which, as shown in the previous item, correspond to the principal part of $\Phi^* f$.

□

Exercise 14.9. Let Q be a non-zero homogeneous polynomial on $t_0, \ldots, t_n$, and $V(Q)$ its zero set, which we consider as a subset in $\mathbb{C}P^n$.

(1) Prove that $\mathbb{C}P^n \setminus V(Q)$ is non-empty.
(2) Prove that $V(Q) \subset \mathbb{C}P^n$ is a set of measure 0.

Proof. (1) This follows from Identity Theorem 13.1. Indeed, if $V(Q)$ was all of $\mathbb{C}^n$, it would vanish on an open set, implying that Q is identically zero.
(2) $V(Q)$ may be decomposed in the sets of regular and singular points. Regular points have submanifold charts, while singular points have measure zero by Sard's theorem.

□

Exercise 14.10. Let $f_1, f_2, \ldots \in \mathcal{O}_n$ be a countable collection of germs, which vanish with multiplicity $k_1, k_2, \ldots$. Prove that there exists a coordinate system $z_1, \ldots, z_n$ such that $\lim_{z_n \to 0} \frac{f_i(0, z_n)}{z_n^{k_i}} \neq 0$ for all i.

Sketch of proof. I don't understand this exercise: evaluating any germ f_i on $(0, z_n)$ will make all terms that have any variable other than z_n vanish. Thus the first term will be a homogeneous polynomial in z_n, which is the principal part of f_i evaluated in $(0, z_n)$. But of course taking quotient by z_n all terms with powers of z_n higher than 1 will vanish, barely leaving the principal part of f_i evaluated at $(0, 1)$. But this works regardless of the coordinate system. □

15. Elementary symmetric polynomials and Newton formula

The main result in this section is to show that the elementary symmetric polynomials can be given in terms of the Newton polynomials. More exactly, as elements of the polynomial ring with rational coefficients and Newton polynomials as indeterminates. In turn, this says that the elementary symmetric polynomials are Weierstrass polynomials as long as the Newton polynomials are holomorphic (which is easy to prove using the Argument Principle, Exercise 10.1). The elementary symmetric polynomials, in the form of a product, are exactly what we get when we factor the zeroes of a holomorphic function as in Equation 10.1.1. So essentially this paragraph says the proof of Weierstrass Preparation Theorem 16.2.

The α_i will eventually play the roles of the zeroes of the holomorphic function, but in this section they are just indeterminates in the ring $\mathbb{Z}[\alpha_1, \ldots, \alpha_n]$. We start by defining three types of polynomials in this ring.

Definition 15.1. Let $e_i \in \mathbb{Z}[\alpha_1, \ldots, \alpha_n]$ be the coefficients of the polynomial

$$t^n + e_1 t^{n-1} + \ldots + e_{n-1} t + e_n := \prod_{i=1}^{n} (t + \alpha_i). \tag{15.1.1}$$

Then e_i are called *elementary symmetric polynomials* on α_i.

Notice that the polynomial in Equation 15.1.1 is an element of the polynomial ring on the indeterminates $t, \alpha_1, \ldots, \alpha_n$.

There are explicit descriptions for the elementary symmetric polynomials. In fact, $e_1 = \sum_i \alpha_i$ (which is the first Newton polynomial p_1 defined below). Also $e_n = \prod_i \alpha_i$, and in general $e_k = \sum_{1 \le i_1 < \ldots < i_k \le n} \alpha_{i_1} \ldots \alpha_{i_k}$.

Definition 15.2. A *Newton polynomial* is $p_j := \sum_{i=1}^{n} \alpha_i^j$.

Definition 15.3. A *complete homogeneous symmetric polynomial* of degree k is h_k obtained as a sum of all homogeneous monomials of degree k, that is, $\alpha_1^k + \ldots + \alpha_n^k + \alpha_1^{k-1}\alpha_2 + \ldots$.

Corresponding to the above definitions we have the generating functions $E(t) := \sum_{i=0}^{n} e_i t^i$, $P(t) := \sum_{i=1}^{\infty} p_i t^i$ and $H(t) := \sum_{i=0}^{\infty} h_i t^i$ which are formal series in $\mathbb{Z}[\alpha_1, \ldots, \alpha_n][\![t]\!]$.

Exercise 15.4. Prove that $H(t) = \prod_{i=1}^{n} \frac{1}{1 - t\alpha_i}$.

Proof. Let us write (possibly as a formal definition)

$$\prod_{i=1}^{n} \frac{1}{1 - t\alpha_i} = \prod_{i=1}^{n} \sum_{k=0}^{\infty} \alpha_i^k t^k.$$

We also write $f_i \overset{\text{ops}}{\longleftrightarrow} \{\alpha_i^k\}_{k=1}^{\infty}$ to mean that f_i is the power series associated to the sequence $\{\alpha_i^k\}_{k=1}^{\infty}$. Then the equation above is the product of the f_i. Then all we have to prove is that

$$\prod_{i=1}^{n} f_i \overset{\text{ops}}{\longleftrightarrow} \left\{ \sum_{i_1 + \ldots + i_n = k} \alpha_1^{i_1} \ldots \alpha_n^{i_n} \right\}_{k=1}^{\infty} = \{h_k\}_{k=0}^{\infty}$$

This is just a generalization of the formula for product of power series for a product of n power series. $\square$

Exercise 15.5. Prove that $E(t) = \prod_{i=1}^{n}(1 + t\alpha_i)$.

Proof. This is just writing $\prod_{i=1}^{n}(1 + t\alpha_i) = \prod_{i=1}^{n} t(\frac{1}{t} + \alpha_i)$ and continue until we get to $E(t)$. □

It follows from the two previous exercises that $H(t)E(-t) = 1$. Using Exercise 15.5 and applying logarithm we obtain that $\frac{E'(-t)}{E(-t)} = -\sum_{i=1}^{n} \frac{\alpha_i}{1-t\alpha_i}$. Expanding this formula as geometric power series we obtain that

$$P(t) = -\frac{E'(-t)}{E(-t)} \tag{15.5.1}$$

Exercise 15.6. Prove that p_i can be expressed as polynomials of e_i (with integer coefficients).

Proof. Using Eq. 15.5.1, and $H(t)E(-t) = 1$, we have

$$P(t) = E'(-t)H(t).$$

Expanding the power series we obtain that the k-th term is

$$p_k = \sum_{i=0}^{k}(-1)^{i+1} i e_i h_{k-i}$$

□

Exercise 15.7. Prove that h_i can be expressed as polynomials of e_i with integer coefficients. Prove that e_i can be expressed as polynomials of h_i with integer coefficients.

Sketch of proof. By $H(t)E(-t) = 1$ we see that $\frac{E'(-t)}{E(-t)} = \frac{H(t)}{H'(t)}$. Both denominators are expressed as power series in α_i. Multiplying by $E(-t)$ as in Exercise 15.5 would let $E'(-t)$ be expressed as a power series in h_i and α_i, while multiplying by $H(t)$ as in Exercise 15.4 would let $H'(t)$ expressed in terms of e_i and α_i. □

Exercise 15.8 (Newton formula)**.** Prove that $ke_k = \sum_{i=1}^{k-1}(-1)^i e_{k-i} p_i$.

Proof. Note that

$$P(t)E(-t) = \left(\sum_{k=0}^{\infty} p_i t^i\right)\left(\sum_{i=0}^{n}(-1)^i e_i t^i\right) = \sum_{k=0}^{\infty}\sum_{i=0}^{k}(-1)^{k-i} e_{k-i} p_i t^k$$

using the product formula of power series, where we define $E(-t)$ as a power series by letting $e_i = 0$ for $i > n$. By Eq. 15.5.1, this equals $E'(-t)$, which we may also see as a power series. Comparing the k-th term yields the result modulo a minus sign. □

Finally,

Exercise 15.9. Prove that e_i are expressed as polynomials on p_i with rational coefficients.

Proof. □

16. Weierstrass preparation theorem

The upshot about Weierstrass preparation theorem is that we would like it if $\mathcal{O}_n$ was an Euclidean domain, i.e. that there was a division algorithm, as in Number Theory Definition 4.1. Probably that's not true because holomorphic functions are infinite series, so we cannot put an Euclidean function like the degree. Instead, we have Weierstrass preparation theorem (which gives a Weierstrass *division* theorem...?)

Before starting let's say the proof. Start with a germ of holomorphic function f. Split the domain as (z', z_n) and regard f as a function of z_n. Then factor its zeroes as in Equation 10.1.1. This gives, for every z', a product of z_n minus the zeroes times a holomorphic function without zeros, namely $u(z', z_n)\prod(z_n - w_i(z'))$. But hey, that product is a polynomial function in the ring $\mathbb{C}[z_n, w_1(z'), \dots, w_n(z')]$; i.e. we shall be done once prove it is in fact a Weierstrass polynomial! By definition, its coefficients are the elementary symmetric polynomials on the $w_i(z')$. But we showed that these can be expressed as polynomials with indeterminates being the Newton polynomials. And hey, the Newton polynomials are holomorphic functions by the Argument principle exercise 10.1. That says that that product of z_n minus the roots is a Weierstrass polynomial, as desired.

Definition 16.1. A *Weierstrass polynomial* is a function $f \in \mathcal{O}_{n-1}[z_n]$, that is, a function which is polynomial in the last variables with coefficients that are analytic functions on the first $n-1$ variables.

The following formulation is found in [Dem]:

Theorem 16.2 (Weierstrass preparation theorem)**.** *Let f be holomorphic on a neighourhood of 0 in $\mathbb{C}^n$, such that $f(0, z_n)/z_n^s$ has a nonzero finite limit at $z_n = 0$. With the "above" choice of r' and r_n, one can write*

$$f(z) = u(z)P(z', z_n)$$

where u is an invertible holomorphic function in a neighbourhood of the polydisk $\overline{\Delta}(r', r_n)$ and P is a Weierstrass polynomial with coefficient functions defined for $|z'| \leq r'$ in $\mathbb{C}^{n-1}$.

Proof. Step 0. First notice that since I will use this theorem to prove that $\mathcal{O}_n$ is a UFD, I will start with an arbitrary function $f \in \mathcal{O}_n$. (Right now it's not clear why we need that $f(0) = 0$.) But then it's straightforward to change coordinates so that f is not identically zero on the z_n-axis: just complete the point where f is nonzero to a basis $\{b_i\}_{i=1}^{n-1}$ and precompose with a linear map sending to $(0, \dots, 0, 1)$ to that point, and the rest of the canonical basis to the corresponding basis vectors b_i. The composition of this linear map with f remains holomorphic and is nonvanishing in the z_n-axis.

(See Exercise 14.10 for a proof that we can suppose that the coordinate change preserves the order of the zero, but again it's not clear why I need that.)

Step 1. Now we define the correct domain. This is necessary for the next step.

Then $f(0, z_n)$ is a holomorphic function with a zero on 0. By the Taylor expansion we know that this zero must be isolated, so that there is a number $r_n > 0$ such that $f(0, z_n) \neq 0$ for $0 < |z_n| \leq r_n$.

Since the circle $\partial\Delta_0(r_n) = \{(0, z_n) : |z_n| = r_n\}$ is compact and $f(0, z_n) \neq 0$ on this circle, we can construct an open neighbourhood of the circle where f is nonzero as follows. For any point of the circle $|z_n| = r_n$ pick a pointed open

neighbourhood where f is nonzero (using continuity). This gives a collection of open neighbourhoods of $\mathbb{C}^n$ covering the circle $\partial\Delta_0(r_n)$, from which we take a finite subcover. Let r' be the minimum radius of these balls. We have that $f(z', z_n) \neq 0$ for $z' < r'$ and $|z_n| < r_n + \varepsilon$ for some small ε.

This means that in the definition of S_k in the next step we are confident that we can integrate the function $\frac{1}{f(z',z_n)}$ along the circle $\partial\Delta_{z'} r_n$ for every z'.

Step 2. Now we apply Argument principle. Consider the function

$$S_k(z') := \frac{1}{2\pi\sqrt{-1}} \int_{\partial\Delta(r_n)} \frac{f'(z', z_n)}{f(z', z_n)} z_n^k dz_n.$$

In Exercise 10.1 we showed that $S_k(z')$ is the sum of the k-th powers of the zeroes of $f(z', -)$ without multiplicities. In particular $S_0(z')$ is an integer: the number of zeroes of $f(z', -)$ on $\Delta_{z'}(r_n)$ (for fixed $z' \in \Delta(r') = \{z' \in \mathbb{C}^{n-1} : |z'| < r'\}$).

The map S_k is holomorphic as a function of z' because it is holomorphic in each of its $n-1$ complex variables. This can be checked by fixing all but one of such variables and splitting the expression in real and imaginary parts. By interchanging partial derivatives and integrals as real functions we conclude that Cauchy-Riemann equations are satisfied by holomorphicity of the integrands. The proof for differentiation of integrals of real functions may be found here.

Moreover, since $S_0(z')$ is an integer, it's constant (as a function of z') by continuity. Thus we have exactly s functions $w_1(z'), \ldots, w_s(z')$ which give the s zeroes of $f(z', -)$ at every $z' \in \Delta(r')$. Now we apply argument principle with multiplication by z^k to obtain $S_k(z') = \sum w_i(z')^k$ (without multiplicities). Again, these maps are holomorphic on z' by Theorem 6.3 since they are defined as integrals and have series expansions. Notice these are by definition the Newton polynomials on $w_i(z')$.

Step 3. By the Newton formula (Exercise 15.8) we know that the elementary symmetric polynomials e_i can be expressed as polynomials of the Newton polynomials p_i. We conclude that the e_i are also holomorphic functions of z'. This shows that

$$\begin{aligned} P(z', z_n) :&= \prod_{i=1}^{k} (z_n - w_i(z')) \\ &= t^n + e_1 t^{n-1} + \ldots + e_{n-1} t + e_n, \end{aligned}$$

where the e_i are the elementary symmetric polynomials, defined exactly by the above relation (see Equation 15.1.1), is a Weiestrass polynomial (because its coefficients are holomorphic functions).

Factoring the zeroes of $f(z', -)$ as in Equation 10.1.1 we obtain

$$f(z', z_n) = \left(\prod_i z_n - w_i(z') \right) h(z', z_n)$$

for a function h that doesn't vanish in $\Delta(r') \times \Delta_0(r_n + \varepsilon)$.

To conclude we need to confirm that h is holomorphic. By our factorization we know that for every fixed z' $h(z', -)$ is holomorphic in z_n. Then we can use Cauchy integral formula 5.3.1 to get

$$h(z', z_n) = \frac{1}{2\pi\sqrt{-1}} \int_{\partial\Delta(r_n)} \frac{f(z', w_n)}{w_n - z_n} dw_n.$$

This is a holomorphic function of z' since for any $z_0' \in \Delta(r')$ the limit

$$\lim_{z' \to z_0'} \frac{\int_{\partial \Delta_{z'}(r_n)} \frac{f(z',w_n)}{w_n - z_n} w d_n - \int_{\Delta_{z_0'}(r_n)} \frac{f(z_0',w_n)}{w_n - z_n} dw_n}{z' - z_0}$$

exists.

□

Recall from Number Theory Lemma **??** that if R is a UFD, then $R[x]$ is a UFD.

Lemma 16.3. *The stalk $\mathcal{O}_n := \mathcal{O}_{\mathbb{C}^n,0}$ is a UFD.*

Proof. By induction on n. For $n = 0$ it is trivial. Suppose $\mathcal{O}_{n-1}$ is a UFD. Then by Gauss' Lemma 6.5, $\mathcal{O}_{n-1}[w]$ is a UFD too. Thus we may express any Weierstrass polynomial g as a product of irreducible elements (uniquely up to multiplication by units).

Let $f \in \mathcal{O}_n$. We want to express f as a product of (unique up to multiplication by units) of irreducible elements. By Weierstrass Preparation Theorem **??** there is a Weierstrass polynomial $g \in \mathcal{O}_n[w]$ and a holomorphic function not vanishing on 0 (i.e. a unit of $\mathcal{O}_n$) such that $f = gh$. By the previous remark g is factored uniquely up to multiplication by units as $g = g_1 \dots g_m$. This shows existence of the factorization.

To prove uniqueness suppose that $f = f_1 \dots f_k$ for some irreducible $f_1, \dots, f_k \in \mathcal{O}_n$. Since f does not vanish in the w axis, neither can each f_i, so that we may decompose each of them as $f_i = g_i' h_i$ by Weierstrass Preparation Theorem. Since f_i is irreducible, it follows that g_i' is irreducible. Then we have that

$$f = gh = \prod g_i' \prod h_i$$

so by uniqueness in Weierstrass Preparation Theorem we conclude that $g = \prod g_i'$, and by uniqueness from the fact that $\mathcal{O}_n[w]$ is a UFD we conclude that g coincides with $\prod g_i'$ up to multiplication by units. □

Here's some material I wrote on the way:

We denote by $B_r(z_1, \dots, z_{n-1})$ the ball of radius r in $\mathbb{C}^{n-1} \subset \mathbb{C}^n$.

Exercise 16.4. Let F be a holomorphic function on a neighbourhood of 0 in $\mathbb{C}^n$, such that $\lim_{z_n \to 0} \frac{F(0,z_n)}{z_n^k} \neq 0, \infty$. Consider the projection map $\Pi : \mathbb{C}^n \to \mathbb{C}^{n-1}$ $(z_1, \dots, z_n) \mapsto (z_1, \dots, z_{n-1})$.

(1) Prove that for an appropriate pair r, r', the resitriction of F to the polydisk $\Delta(n-1, 1) := B_r(z_1, \dots, z_{n-1}) \times \Delta_{r'}(z_n)$ nowhere vanishes on the set $\Pi^{-1}(\partial \Delta_{r'}(z_n))$.
(2) Prove that in this case the resitriction of F to this polydisk has precisely k zeroes $\alpha_1, \dots, \alpha_k$ on each fiber of Π.
(3) Prove that $\sum_{i=1}^k \alpha_i^d$ is a holomorphic function on $B_r(z_1, \dots, z_{n-1})$.
(4) Prove that any elementary symmetric polynomial on α_i gives a holomorphic function on $B_r(z_1, \dots, z_{n-1})$.

Proof. (1) Suppose that for every (r, r') there are points $z_1 \in B_r(z_1, \dots, z_{n-1})$ and $z_n \in \partial \Delta_{r'}(z_n)$ where F vanishes. Then we obtain a convergent sequence where F vanishes, and by Identity Principle 13.1 we conclude that F mus be identically zero.

(2) This is just applying Argument principle, i.e. the integral $\frac{1}{2\pi\sqrt{-1}}\int_{\partial\Delta}\frac{F'(y_0,z)}{F(y_0,z)}dz$ equals the number of zeroes. Since it is a continuous (holomorphic?) function on Δ_y and integer valued, it must be constant.

The condition that $\lim_{z_n\to 0}\frac{F(0,z_n)}{z_n^k}\neq 0,\infty$ means that the fiber at 0 has a zero of order k:

$$\frac{F(0,z_n)}{z_n^k}=\frac{F(0)}{z_n^k}+F'(0)\frac{z_n}{z_n^k}+\ldots+\frac{F^{(k)}(0)}{k!}\frac{z_n^k}{z_n^k}+\ldots$$

Thus, on other fibers we can have more zeroes, but without multiplicities they are always k.

(3)

□

Exercise 16.5 (Weierstrass preparation theorem)**.** Let F be an analytic function in a neighbourhood of 0 in $\mathbb{C}^n$, such that $\lim_{z_n\to 0}\neq 0,\infty$. Consider the projection map $\Pi:\mathbb{C}^n\to\mathbb{C}^{n-1}$, $(z_1,\ldots,z_n)\mapsto(z_1,\ldots,z_{n-1})$, and let $P(z_n)\in\mathcal{O}_{n-1}[z_n]$ be the Weierstrass polynomial given by $P(z_n)=\sum_{i=0}^k e_i z_n^i$, where e_i are the elementary symmetric polynomials on the zeros $\alpha_1,\ldots,\alpha_k$ defined in the previous exercise. Prove that $F=P(z_n)u$, where u is a germ of an invertible holomorphic function.

Proof. For every y_0 we can write

$$F(y_0,z_n)=(z_n-\alpha_1)\ldots(z_n-\alpha_n)h(z_n)$$

where implicitly all α_i and h depend on y_0. The product of $(z_n-\alpha_i)$ is the definition of P. Varying y_0 we obtain the result. □

Also [GH78] formulation:

As I recall the following is [GH78] formulation of the theorem:

Theorem 16.6 (Weierstrass preparation theorem)**.** *If $f:U\subset\mathbb{C}^n\to\mathbb{C}$ is holomorphic and f is not identically zero in the coordinate axis $z_n:=w$, there is a unique germ of a monic Weierstrass polynomial g whose coefficients are holomorphic functions on the first $n-1$ variables and a germ of a holomorphic function h with $h(0)\neq 0$ (i.e. h is a unit of $\mathcal{O}_n$) such that $f=gh$.*

References

[Ahl79] L. Ahlfors, *Complex analysis: An introduction to the theory of analytic functions of one complex variable*, McGraw-Hill Education, 1979.

[Dem] Jean-Pierre Demailly, *Complex analytic and differential geometry.*

[GH78] Phillip Griffiths and Joseph Harris, *Principles of algebraic geometry*, Pure and Applied Mathematics. A Wiley-Interscience Publication., John Wiley & Sons, New York, 1978.

[Lee12] John M. Lee, *Introduction to smooth manifolds*, second edition ed., Springer New York, New York, NY, 2012.

[Lee24] J.M. Lee, *Introduction to complex manifolds*, Graduate Studies in Mathematics, American Mathematical Society, 2024.